The Media Connectivity Association



OVERVIEW AND UPDATES · CONTENT DELIVERY AND STREAMING

Video Operation Points

Video capabilities and operation points of 3GPP TS 26.265, and the conformance validator for them

3GPP

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About the project

What 3GPP TS 26.265 defines, and the part the validator comes from

Video capabilities

Part of 3GPP TS 26.265

Service-independent video decoding and encoding capabilities, which 3GPP service specifications and third-party services may reference.

Operation points

Part of 3GPP TS 26.265

Combinations of a video coding format and a representation format, with requirements for the bitstream and for the receiver.

Conformance

Part of 3GPP TS 26.265

The conformance functions, among them the conformance validator that tests whether a bitstream conforms to an operation point.

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Operation points

The seven operation points of TS 26.265, grouped by video format

3GPP-HD

AVC and HEVC

3GPP-AVC-HD: AVC High Progressive Profile, Level 4.0. 3GPP-HEVC-HD: HEVC Main 10 Profile, Main Tier, Level 4.1.

3GPP-HDR

HEVC, HD and UHD

HEVC Main 10 Profile, Main Tier: 3GPP-HEVC-HD-HDR at Level 4.1 and 3GPP-HEVC-UHD-HDR at Level 5.1.

3GPP-STEREO

HEVC and MV-HEVC

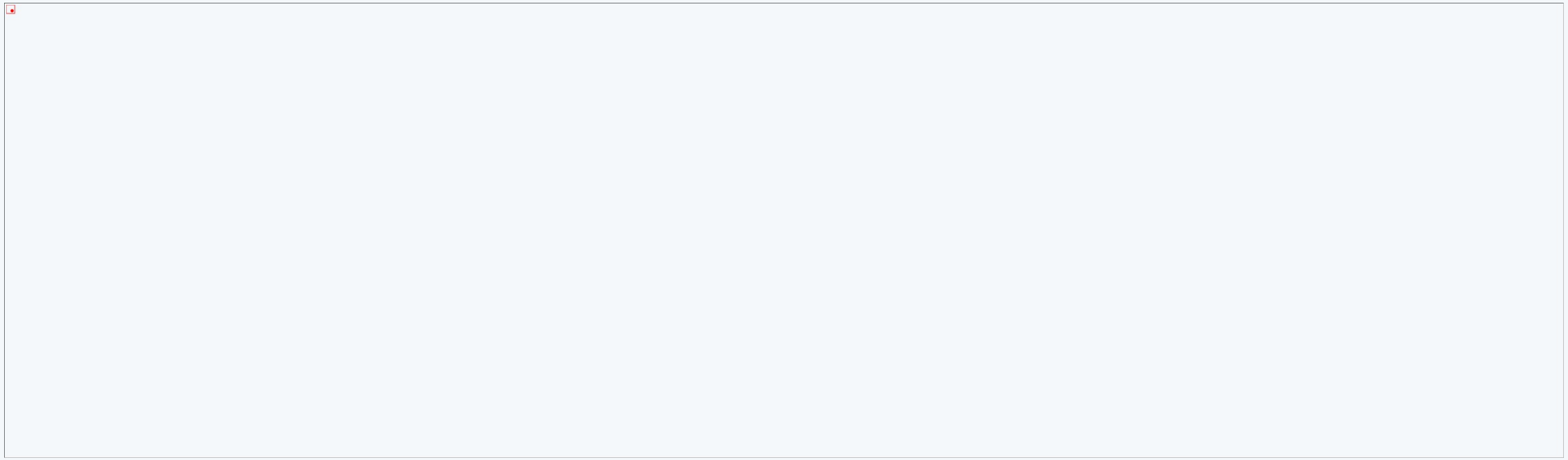
3GPP-HEVC-Stereo: HEVC Main 10 Profile, Main Tier, Level 5.2, frame packed. 3GPP-MV-HEVC-Main-Stereo and 3GPP-MV-HEVC-Ext-Stereo: MV-HEVC, two layers.

An operation point is a collection of discrete combinations of video representation formats: spatial and temporal resolutions, colour mapping, transfer functions and the encoding format.

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Reference architecture

From a video signal to rendering, in the specification



Redrawn from TS 26.265. The APIs are not normatively specified: they serve as an example reference to configure encoders and decoders.

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Conformance functions

What the conformance part of TS 26.265 provides

Test material	Validator	Conforming decoders	Conformance bitstreams
Annotated video test signals, and reference encoders with configurations.	Tests bitstreams that claim to conform, and reports which test assertions are not fulfilled.	Return a report and an MD5.	Give a decoder under test a bitstream, and the decoder conformance test its information.

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How the validator works

It parses a bitstream to XML, then checks the XML against schemas

1 Parse bitstream	2 Generate XML	3 Load XSD schemas	4 Validate
Reads the NAL units of the AVC or HEVC bitstream and parses their headers and parameters.	Creates an XML element for each NAL unit, with its parameters nested by codec structure.	Reads the operation point and loads the XSD 1.1 schemas specified for it.	Runs the XSD validation, extracts which assertions pass or fail and writes the report.

From the repository README. A "shall" requirement that fails is reported as an error, a "should" as a warning.

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How it grew

Video operation points before and in TS 26.265, release by release

Release 15 to 18

Earlier operation points in other specifications: HDR and VR profiles (TS 26.116, TS 26.118), an 8K TV operation point (TS 26.116, TS 26.511).

Release 19

TS 26.265 is created by the VOPS work item: operation points and capabilities consolidated, with MV-HEVC stereo operation points and conformance assets.

Release 20

Release 20 (work plan of 28 September 2026): no work item lists TS 26.265. The study FS_AVFOPS_MED (TR 26.966) is listed; its relation to TS 26.265 is not stated.

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What is still open

Points the specification leaves for further study, which bound the validator

System integration

Definitions for AVC and HEVC

The specification states that the definitions for AVC and for HEVC are for further study.

Encoder API

Mapping of operation points

The mapping of operation points to an encoder API is stated as work for further study.

Validator report

Semantics and format

The detailed semantics and the format of the report are for further study, so the JSON report is the validator's own.

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Repositories

The conformance validator, from the description of its repository

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE	
vops-conformance-validator	Checks AVC, HEVC and MV-HEVC bitstreams against the operation points of TS 26.265; some requirements only partly	AVC, HEVC, MV-HEVC, XSD 1.1 schemas, JSON report	none yet	GitHub

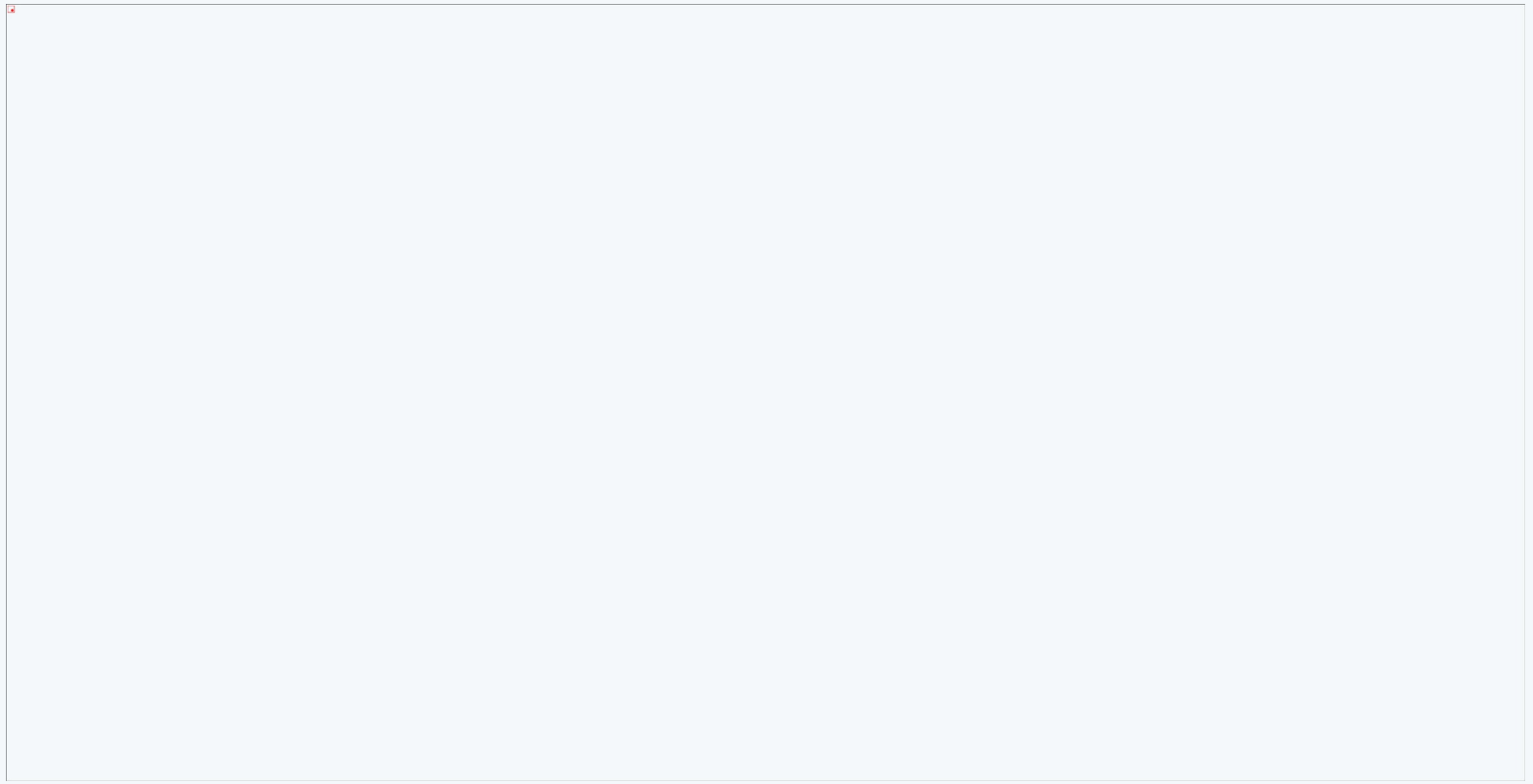
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released no release yet

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Repositories in the architecture

The Reference Tools on the specification figure, as the figure generator draws them



PROVIDED BY THE REFERENCE TOOLS

1 **vops-conformance-validator**

Only the Conformance Validator is mapped to a repository.

WHERE WE STAND

Video Operation Points

BROUGHT TO STUDY	ON-GOING	OUTCOMES
Keeping the validator in step with the specification	Porting of VOPS Validator from 3GPP Fixes identified on verdicts	Porting of Validator from 3GPP under the umbrella of the Media Connectivity Software Accelerator

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